



Government of Pakistan
National Vocational & Technical Training Commission (NAVTTTC)
Plot No. 38 Kirthar Rd, H-9/4, Islamabad.



Roll no: 1155824

Name: Alisha Khan

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Trade: Artificial Intelligence (Machine Learning, Deep Learning, Communication)

Test Date & Time: 22/05/2026 02:00 PM

Test Center Name: Institute of Business Administration
Karachi

Examiner Name: Farhan /

Signatures

Candidate Signature

Examiner Signature & Stamp

Principal Signature

K.M Zarar

Instructions:

Please read instructions carefully before attempting paper.

- 1) All questions are compulsory.
- 2) The question paper consists of 21 questions divided into two sections A & B. Section A comprise of 20 MCQs of 1 mark each. Section B comprises of 1 practical task having 80 marks.
- 3) Mobile Phone not allowed.
- 4) Use separate paper for rough work.

Section A - Theory (Total Marks: 20, Time: 30 Minutes)

1- In machine learning, what type of problem involves predicting a continuous numerical output?

- ☒ (a) Regression
- (b) Classification
- (c) Reinforcement learning
- (d) Clustering

2- In natural language processing, what mechanism is used to selectively focus on different parts of input sequences while making predictions, enabling the model to assign varying degrees of importance to different elements?

- (a) Bag-of-Words
- ☒ (b) Attention Mechanism
- (c) Named Entity Recognition
- (d) Sequence-to-Sequence Model

3- Which command is used to change the permissions of a file in Unix-like system

- (a) touch
- (b) kill
- ☒ (c) chmod
- (d) ps

4- In programming, what is the purpose of a function?

- ☒ (a) To organize code into reusable blocks
- (b) To create loops
- (c) To print output
- (d) To store data

5- In machine learning, which algorithm creates a tree-like model of decisions and their possible consequences, where each internal node represents a decision based on an attribute, and each leaf node represents an outcome?

- (a) Linear Regression
- ☒ (b) Decision Tree
- (c) K-Means Clustering
- (d) Support Vector Machine (SVM)

6- Which Seaborn function is used to create a bar plot of categorical data?

- (a) sns.catplot()
- ☒ (b) sns.barplot()
- (c) sns.scatterplot()
- (d) sns.lineplot()

7- What does broadcasting in Numpy allow you to do?

- (a) Convert arrays to different data types
- ☒ (b) Perform element-wise operations on arrays with different shapes
- (c) Combine arrays into a single array
- (d) Resize arrays to match the largest array

8- In NLP, what is the process of reducing words to their base or dictionary form while ensuring they are still valid words?

- ☒ (a) Lemmatization
- (b) Tokenization
- (c) Chunking
- (d) Stemming

9- What are the fundamental building blocks of programming that manipulate data?

- ☒ (a) Operators
- (b) Keywords
- (c) Variables
- (d) Functions

10- How does the Anomaly Detector service in Microsoft Azure work?

- (a) It identifies abnormalities based on statistical analysis
- (b) It compares data to pre-defined patterns
- (c) It relies on user-defined rules to detect anomalies
- ☒ (d) It uses machine learning to predict anomalies

NOTICE



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11- What is the primary purpose of string operations in programming?

- (a) To generate random numbers
- (b) To control robotic movements
- (c) To create complex mathematical equations
- ☒ (d) To manipulate and process text

12- In object-oriented programming, what is multilevel inheritance?

- (a) Inheriting from multiple parent classes
- (b) Inheriting from a child class
- ☒ (c) Inheriting from a grandparent class
- (d) Inheriting from a single parent class

13- Which Numpy function is used to perform element-wise addition of two arrays?

- ☒ (a) numpy.add()
- (b) numpy.append()
- (c) numpy.sum()
- (d) numpy.plus()

14- In probability theory, what type of probability considers the likelihood of events occurring together?

- ☒ (a) Joint probability
- (b) Conditional probability
- (c) Independent probability
- (d) Marginal probability

15- Which service can you use to analyze images and extract text from them in Microsoft Azure?

- (a) Anomaly Detector
- (b) Personalizer
- (c) Metrics Advisor
- ☒ (d) Content Moderator

16- What is the purpose of "magic functions" or "dunder methods" in Python?

- (a) Enforcing strict naming conventions
- (b) Defining private methods in a class
- (c) Creating unique variable names
- ☒ (d) Overloading operators and providing special behavior

17- What is the advantage of using functions in programming?

- (a) They allow you to define new data types
- (b) They help you optimize memory usage
- (c) They allow you to create GUI interfaces
- ☒ (d) They allow you to reuse code and improve code organization



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18- In logistic regression, what type of output does the model produce?

- ☒ (a) Probability scores
- (b) Continuous numerical values
- (c) Categorical labels
- (d) Clusters of data points

19- What is the primary purpose of Artificial Intelligence (AI)?

- (a) To simulate human emotions.
- (b) To play board games.
- (c) To create physical robots only.
- ☒ (d) To perform tasks without any human intervention.

20- What is the primary purpose of using nested loops in programming?

- (a) To reduce code complexity
- (b) To create recursive functions
- ☒ (c) To execute a loop inside another loop
- (d) To improve code readability

__The End__